



Mutah University, Karak Jordan
Faculty of Information Technology

TECHNO-VALLEY

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CERTIFICATE

It is hereby certified that the project titled ***TECHNO-VALLEY***, submitted by undersigned, in partial fulfillment of the award of the degree of “Bachelor in Computer science” embodies original work done by them under my supervision.

All the analysis, design and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

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ABSTRACT

Techno-Valley is an innovative online platform specifically designed to serve the academic and collaborative needs of IT faculties in Jordanian universities. The platform provides three distinct user roles: visitors, who can browse content; students, who can browse and publish content using their university IDs; and experts, typically professors, who are verified through their university credentials.

The platform focuses on creating a structured environment for publishing academic resources, supporting file types such as text and PDF, and enabling advanced content discoverability through a hashtag-based search system. By fostering an ecosystem of collaboration, Techno-Valley ensures verified and credible content, empowering students and academics to connect, share knowledge, and enhance their professional and educational experiences.

This project aims to address gaps in existing systems by providing a user-friendly, role-specific solution tailored to the IT academic community, emphasizing accessibility, credibility, and relevance to modern academic needs. The expected outcomes include a responsive and user-friendly interface, a verified system for academic content, and a streamlined process for academic collaboration.

ACKNOWLEDGEMENTS

Lastly, we wish to convey our sincere thanks and deep appreciation to our supervisor, Dr. Khalid Awad for investing time in reviewing our work and providing us the needed guidance and support. We are also immensely grateful to IT college, Mutah University.

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ABBREVIATIONS

Student(stud): - someone who is actively engaged in learning, typically at an educational institution like college or university.

Professor(prof): - is an academic title used to describe a teacher or instructor at a college or university level, typically someone with advanced expertise in a particular subject. Professors are responsible for teaching, conducting research, and contributing to the academic community.

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CHAPTER 1: INTRODUCTION

1.1 Overview

Our website is designed to assist students in searching for information efficiently and in a customized manner without relying on secondary applications. Dedicated users, such as university students and professors from IT faculties, can upload content, ensuring academic credibility. The hashtag feature facilitates efficient information searching and filtering, making the platform highly user-friendly.

1.2 Project Motivation

- **Why this project?** The search for information is a cornerstone of modern education. Simplifying this process represents a leap in educational development, saving time, effort, and costs.
- **Why is it important?**
 - Reduces search effort through a streamlined process.
 - Saves time by making information quickly accessible.
 - Minimizes costs by eliminating the need for external resources.
- **What is new?**
 - Verified users: Only Jordanian university members can upload content, ensuring credibility.
 - Specialized features for IT colleges, such as academic hashtags and content filtering.

1.3 Problem Statement

- **Security:** Ensuring all uploaded files are verified.
- **Monitoring:** Confirming content relevance to associated hashtags.
- **Optimization:** Creating a responsive and smooth website experience.
- **Authentication:** Limiting uploads to users from Jordanian universities.

1.4 Project Aim and Objectives

The primary goal of this project is to develop a user-friendly, secure, and reliable platform for IT students and professors in Jordanian universities. Specific objectives include:

- Designing a responsive website with flexible functionality.
- Establishing a trusted academic environment with verified users.
- Providing advanced search capabilities using hashtags.

1.5 Project Scope

The platform will include:

- Verified content uploads.
- Advanced search with hashtag-based categorization.
- Role-based access control for visitors, students, and experts.
- Academic collaboration features specific to IT faculties.

The Pilot Study:

As part of defining the project scope, a survey was conducted to gather data about the target audience. The results showed that most users are students from technical disciplines such as Computer Science and Information Systems. The most sought-after features included search by hashtags and file sharing, emphasizing the importance of integrating these functionalities into the system.

We did a graphical analysis of the data as shown:

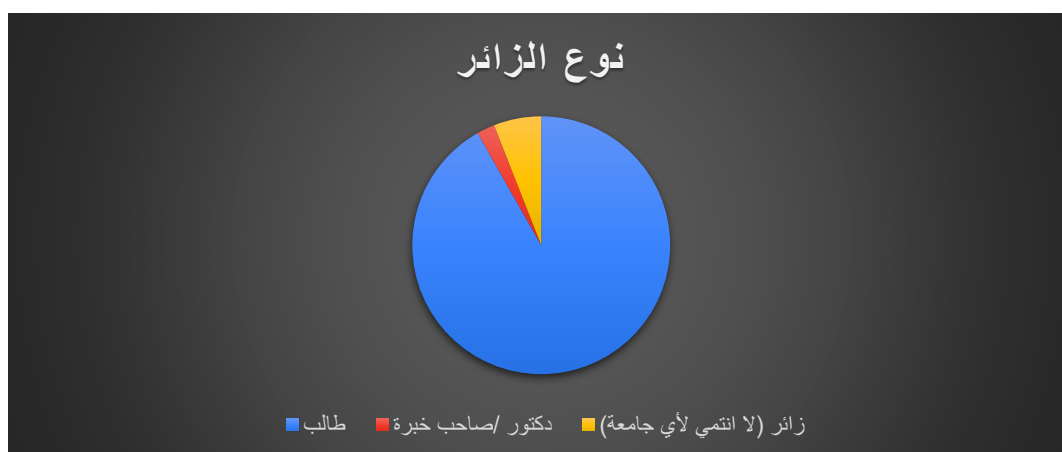


Figure 1 (1.1): نوع الزائر

هل سبق لك استخدام منصات تعليمية أو مواقع للتعليم؟



Figure 2 (1.2): تجارب سابقة

كم مرة تحتاج إلى البحث عن ملفات أو مواد متعلقة بدراستك في الأسبوع؟



Figure 3 (1.3): معدل البحث

هل ترى أن تقييد النشر للطلاب وأعضاء هيئة التدريس فقط فكرة جيدة؟



Figure 4 (1.4): تقييد النشر

1.6 Project Software and Hardware Requirements

- Software Requirements:

Operating System: Windows, macOS, or Linux.

Frontend: HTML, CSS, JavaScript.

Backend: Kotlin with spring boot framework.

Database: MySQL.

-Hardware Requirements:

Developer Machines: Laptops or desktop computers with sufficient processing power, memory (RAM), and storage space for development tasks.

Internet Connectivity: For accessing online resources, collaboration, and code repositories.

1.7 Project Limitations

1. **Dependence on Internet Connectivity:** The platform requires a stable internet connection for users to access content and perform operations.
2. **Content Moderation Challenges:** Ensuring that all uploaded content adheres to academic standards may require additional resources for moderation.
3. **Scalability Issues:** The system might face performance challenges as the number of users and content grows.
4. **Limited File Formats:** Currently, the platform only supports text and PDF files, which may restrict the variety of academic resources.
5. **Verification Process:** Dependence on university systems for verifying user credentials could introduce delays in account activation.
6. **Language Support:** Initially, the platform may only support English, limiting accessibility for users preferring other languages.
7. **Mobile Optimization:** While the platform is responsive, it may not provide the same seamless experience as a dedicated mobile app.
8. **Resource Constraints:** Budget and team size may limit the inclusion of advanced features like AI-powered search or video uploads in the initial release.
9. **Maintenance Requirements:** Regular updates and fixes will be needed to address bugs and enhance system functionality.

1.8 Project Expected Output

- 1- A responsive, user-friendly interface.
- 2- Verified, reliable academic content.
- 3- Advanced search capabilities for streamlined collaboration.

Table 1 (1.1): Teamwork

name	Work on
talal	Chapter 1, chapter 2, chapter 3
Ibrahim	Chapter 3, prototype (frontend) , Chapter 4(User interface design & sequence diagram)
Asmaeel	Chapter 3, chapter 4

1.9 Project Schedule

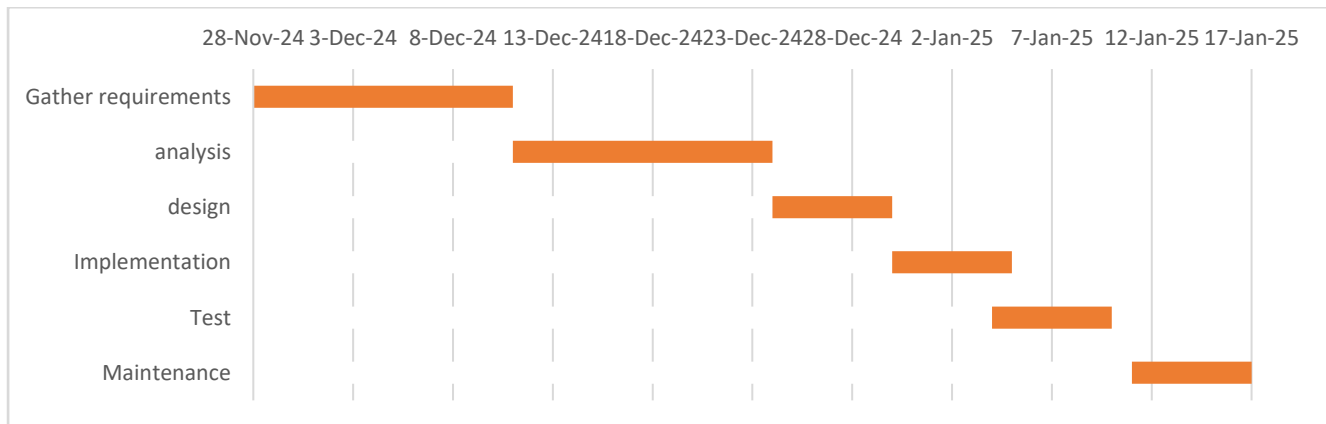


Figure 5 (1.5): A Gantt Chart

1.10 Project, product, and schedule risks

Scope creep: Uncontrolled changes to project requirements.

Technological Hurdles: Unfamiliar or unreliable software/hardware.

1.11 Report Organization

The rest of the report is organized as follows.

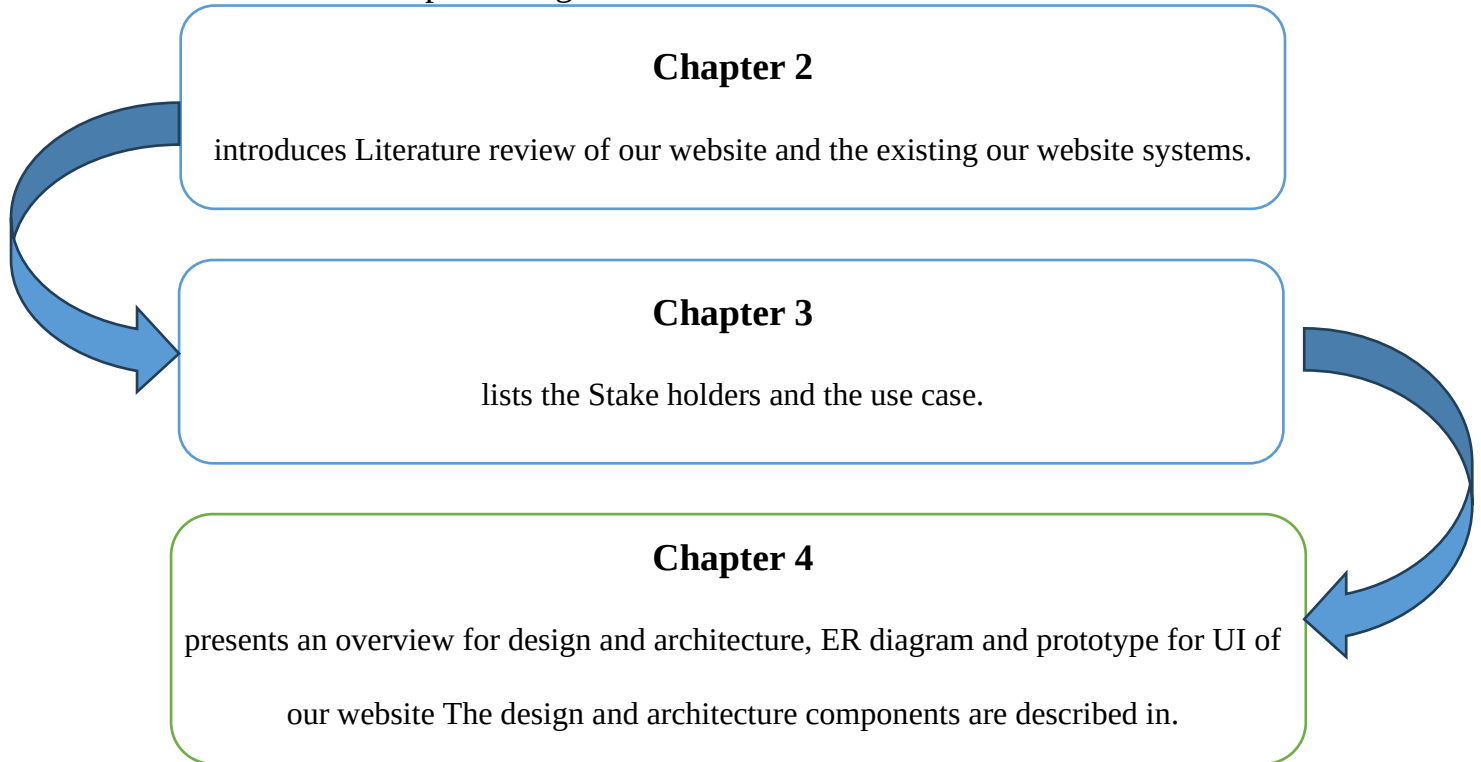


Figure 6 (1.6): Report Organization

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The literature review provides a foundation for understanding the existing systems and technologies related to "Techno-Valley." This chapter explores prior research and platforms, highlighting their functionalities, limitations, and relevance to the project's objectives. The review focuses on online academic platforms, tagging systems, and role-based user access, showcasing the need for a specialized platform for IT colleges in Jordanian universities. The goal is to position "Techno-Valley" as a tailored solution that addresses the gaps in existing platforms.

2.2 Existing Systems

Several platforms and systems support academic collaboration and content sharing, including:

Learning Management Systems (LMS): Platforms like Moodle and Blackboard are widely used for managing educational content. While they offer robust tools for course management, they are not designed for public accessibility or collaboration between students and professors outside specific courses.

Online Forums and Knowledge Bases: Websites like Stack Overflow and Quora provide forums for knowledge sharing. However, these platforms lack academic verification, which is critical for university-level content.

Social Media Platforms: Platforms such as Facebook and LinkedIn allow for networking and content sharing but do not support structured academic content or controlled access based on university credentials.

2.3 Overall Problems of Existing Systems

Existing systems face several challenges that hinder their suitability for "Techno-Valley's" objectives:

- **Verification:** Most platforms lack mechanisms to verify academic credentials, leading to unmoderated content.
- **Target Audience:** General platforms do not focus on the specific needs of IT colleges, limiting their relevance.
- **Content Searchability:** Few systems provide advanced tagging and search features for academic topics.

- **Limited Role Differentiation:** User roles in existing platforms are often not tailored to specific academic structures, such as distinguishing between students, professors, and general visitors.

2.4 Overall Solution Approach

"Techno-Valley" addresses these challenges through:

- **Role-Based Access Control:** Implementing distinct access levels for visitors, students, and experts (professors) based on university credentials.
- **Academic Verification:** Ensuring that only verified users can post content, fostering a credible and professional environment.
- **Tagging System:** Enabling users to categorize content using hashtags (#), improving searchability and content organization.
- **Focus on IT Colleges:** Tailoring the platform's features to the specific needs of IT faculties in Jordanian universities, ensuring relevance and usability.

CHAPTER 3: REQUIREMENT COLLECTION/ENGINEERING AND ANALYSIS

3.1 Stakeholders

Students: The primary users of the platform, responsible for uploading and accessing academic content. Success for them is having an intuitive interface, verified resources, and effective search capabilities.

Experts (Professors): Professors will ensure the credibility of the shared content. Their role includes providing advanced academic materials and engaging with students. Success is defined as smooth interaction and reliable content management tools.

Visitors: General users who can browse and explore the platform without contributing content. Their success is a seamless browsing experience.

Administrators: Manage user accounts, monitor uploaded content, and ensure the system runs efficiently. Success for them involves robust tools for user and content moderation.

3.2.1 Use Case Diagram:

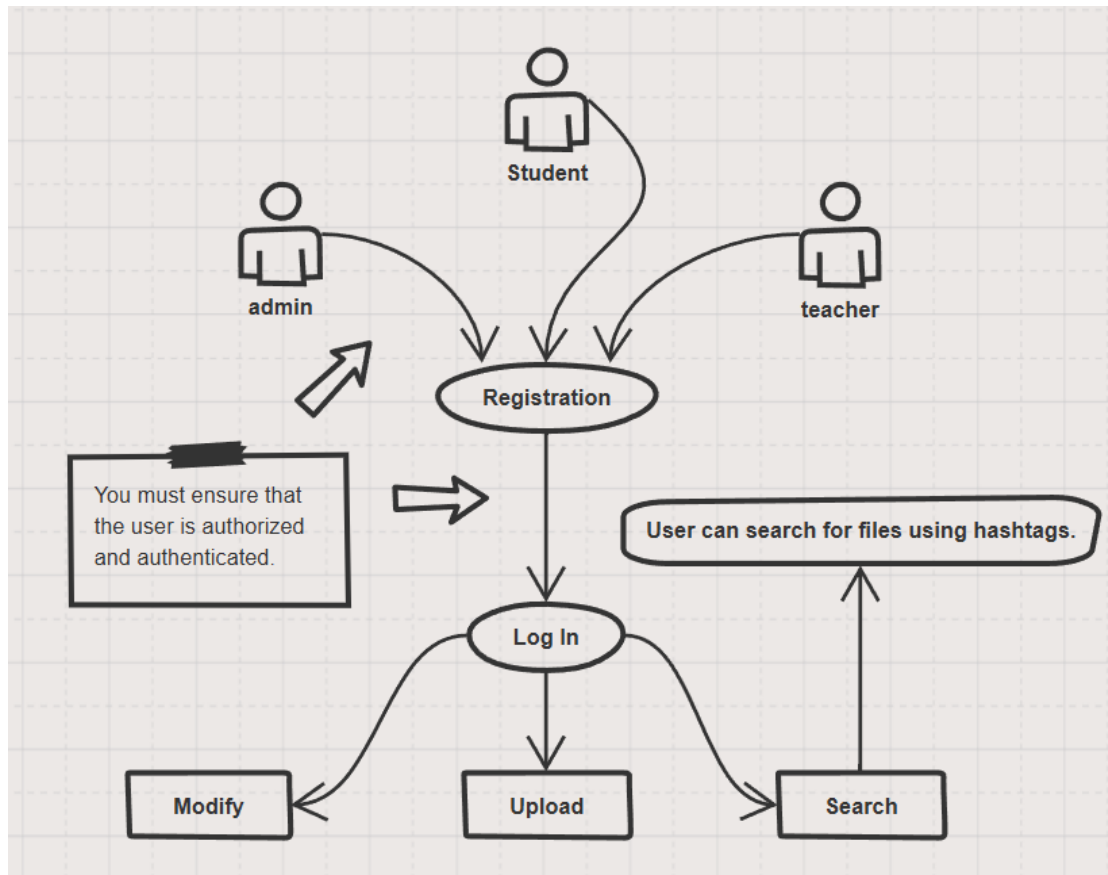


Figure 7 (3.1): Use Case Diagram

3.2.2 Use Case Section

Table 2 (3.1): Registration

key	Value
Use case name:	Registration.
Actors:	Students, Professors, Teachers.
Preconditions:	None
Basic Flow:	1. User visits the registration page. 2. Enters personal and university ID details. 3. The system validates the information. 4. Confirmation email is sent to complete registration.
Postconditions:	User account is created and stored in the database.

Table 3 (3.2): Login

key	Value
Use case name:	Login.
Actors:	All users.
Preconditions:	User must be registered.
Basic Flow:	1. User provides login credentials. 2. System verifies credentials. 3. Grants access to respective role-based functionalities.
Postconditions:	User is directed to their dashboard.

Table 4 (3.3): Upload Content

key	Value
Use case name:	Upload Content.
Actors:	Students, Professors, Teachers.
Preconditions:	Login is required.
Basic Flow:	1. User uploads a file (text or PDF). 2. Adds a descriptive title and relevant hashtags. 3. The system verifies the content for format and completeness. 4. Content is published.
Postconditions:	Content becomes searchable and accessible to other users.

Table 5 (3.4): Search Content

key	Value
Use case name:	Search Content.
Actors:	All users.
Preconditions:	None (There is an ability to add a visitor to the site) not sure!
Basic Flow:	1. User enters a keyword or hashtag in the search bar. 2. System retrieves matching results. 3. Displays results sorted by relevance and date.
Postconditions:	User views the desired academic material.

Table 6 (3.5): Content Moderation

key	Value
Use case name:	Content Moderation.
Actors:	Administrators.
Preconditions:	Login is required.
Basic Flow:	1. Administrator reviews flagged or reported content. 2. Takes appropriate action (approve, edit, or remove).
Postconditions:	Content meets platform standards.

3.2.3 sequence diagram:

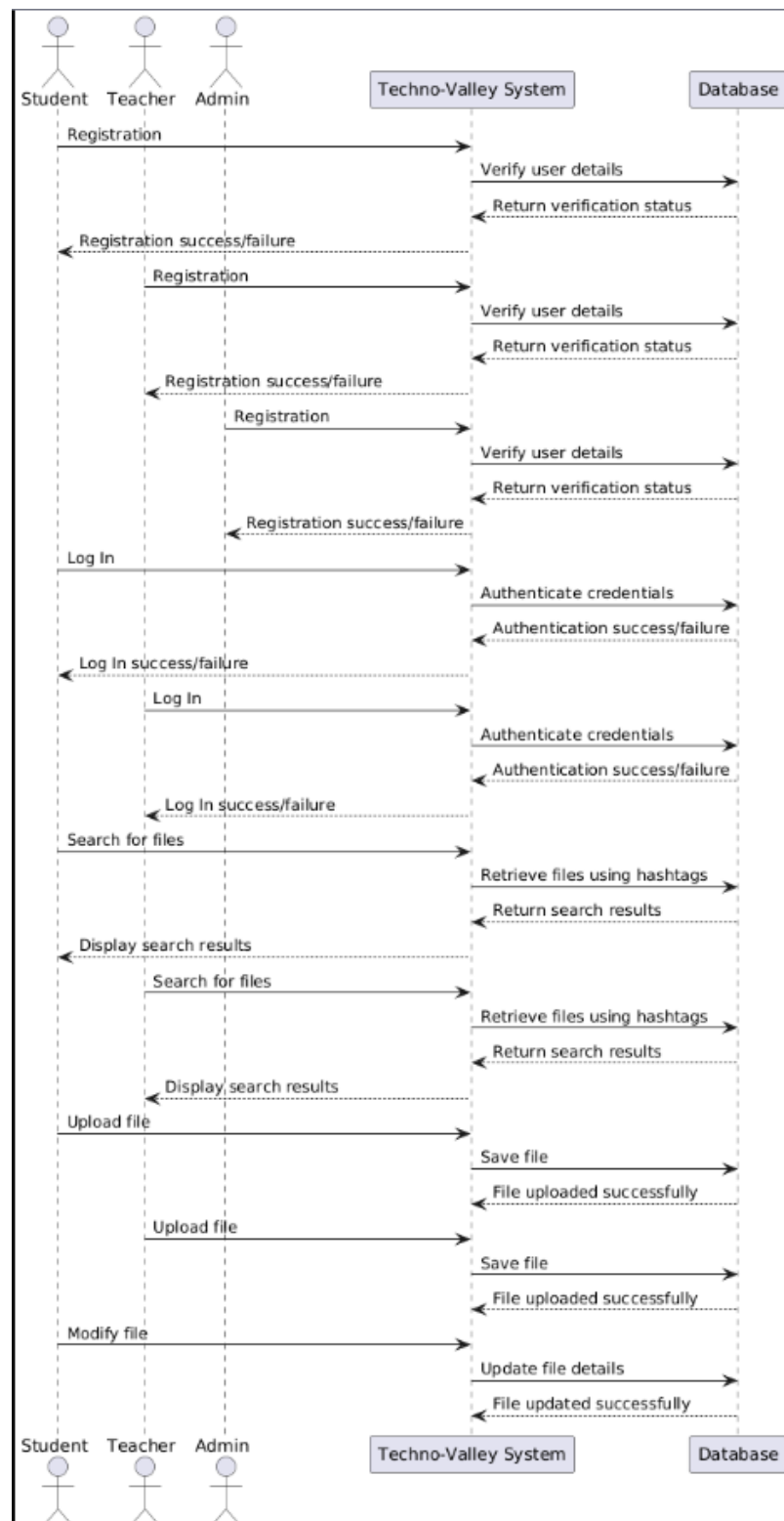


Figure 8 (3.2): sequence diagram

3.3 Non-Functional User Requirements:

Execution Qualities:

- **Performance:**
 - The system should respond to user actions within 2 seconds for tasks such as login and search.
- **Security:**
 - Ensure sensitive user data, including university credentials, is encrypted.
 - Protect the system from unauthorized access.
- **Scalability:**
 - The system must handle up to 1,000 concurrent users without degradation in performance.

Usability:

- **Learnability:**
 - New users should be able to navigate the system efficiently within 10 minutes.
- **Error Prevention:**
 - Provide meaningful error messages and validation prompts.

Evolution Qualities:

- **Maintainability:**
 - The system's codebase must be modular and well-documented.
- **Extensibility:**
 - Designed to accommodate future features such as video content uploads.
- **Testability:**
 - Incorporate robust testing frameworks to validate system integrity.

3.4 Constraints

- **Regulatory:**

- Comply with university policies for online platforms.

- **Technical:**

- Ensure compatibility with major browsers and devices.
- Store files securely with backup mechanisms.

- **Operational:**

The platform should remain accessible 24/7 with minimal downtime

CHAPTER 4: ARCHITECTURE AND DESIGN

4.1 Introduction

The architecture and design of Techno-Valley form the backbone of the platform, ensuring it meets user requirements efficiently. This section focuses on the structural and functional components of the system, detailing how the architecture is built for scalability, security, and ease of use. The database design establishes robust relationships between core entities, while the user interface design prioritizes accessibility and responsiveness to cater to diverse user roles.

4.2 Overview:

Basic Architecture:

1. **Client-Side (Frontend) Frameworks:**
 - Use **HTML, CSS, and JavaScript** to build an intuitive and responsive interface.
 - Ensure compatibility across desktops, tablets, and smartphones.
2. **Server-Side (Backend) Frameworks:**
 - Use **Spring Boot with Kotlin** for the backend to handle logic, data processing, and API services.
3. **Database:**
 - Use **My SQL Server** to manage user data, posts, files, and hashtags efficiently.

Key Features:

- **User Management:**
 - **Sign Up/Login:** Secure authentication and account creation for students and teachers.
 - **User Roles:** Define distinct privileges for teachers and students.
 - Teachers can document their roles (Trainer, Teacher, or Instructor).
 - Students can upload and manage their educational files.
- **File Sharing and Post Management:**
 - **Upload and Download:** Users can upload files (keys, educational resources) and download shared files.
 - **Hashtags:** Organize and search posts by hashtags.
 - **Personal Pages:** Each user has a dedicated page to showcase their posts, files, and information.
- **Search Functionality:**
 - **Search by Hashtags:** Quickly find files and resources.
 - **Search for Users:** Locate profiles based on user attributes.

User-Friendly Design:

1. **Navigation:**
 - Simple and intuitive menus for effortless browsing.
 - Dedicated sections for posts, search, and profile management.
2. **Accessibility:**
 - Ensure usability for all, including those with assistive technologies.

Scalability and Performance:

- Design the system to handle a growing number of users.
- Implement caching to improve search and data retrieval performance.

Design Principles:

- **User Experience (UX):**
 - Easy Navigation: Prioritize clarity and ease of use.
 - Accessibility: Follow standards for universal usability.
- **User Interface (UI):**
 - Clean Design: Use modern aesthetics with professional typography and spacing.
 - Responsive Layout: Ensure a seamless experience across devices.

4.3 Data Base Design & ER Diagram:

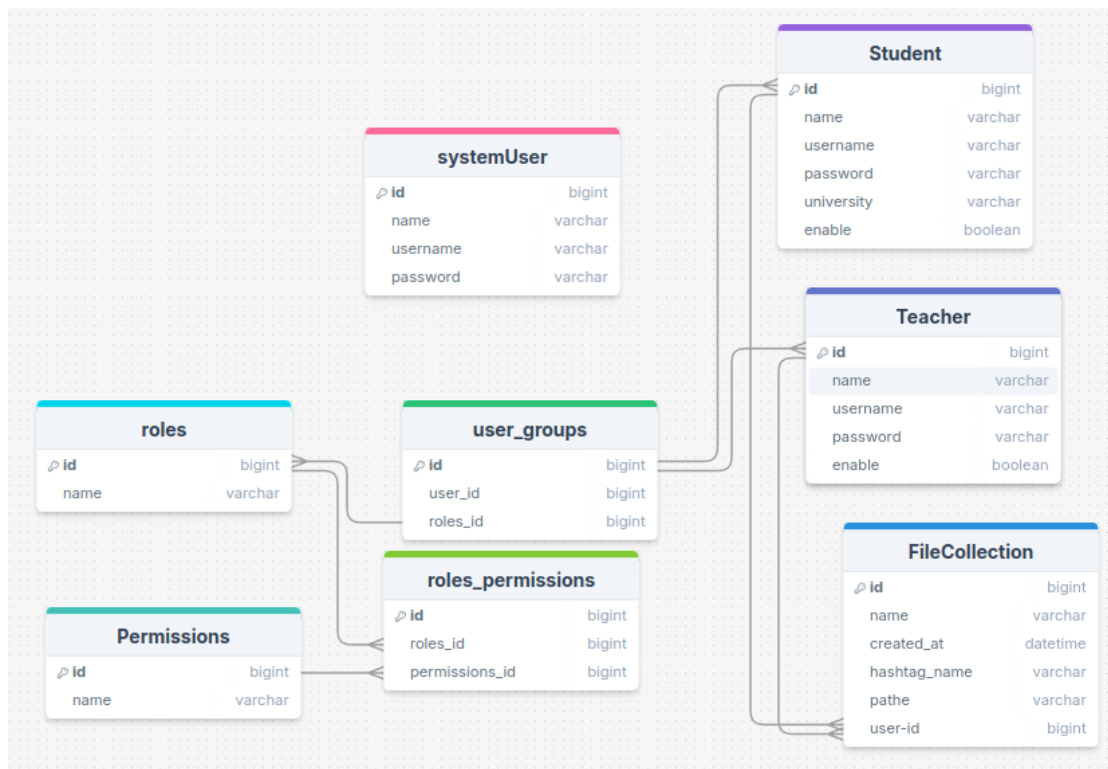


Figure 9 (4.1): Data Base Design

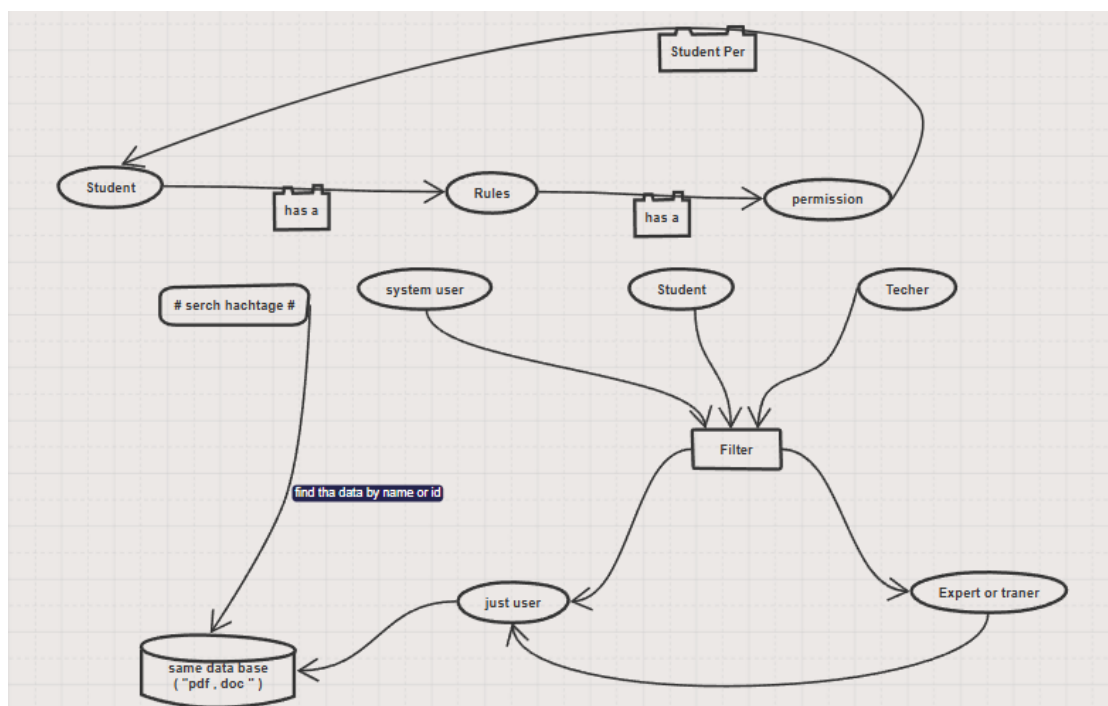


Figure 10 (4.2): Data Base explain

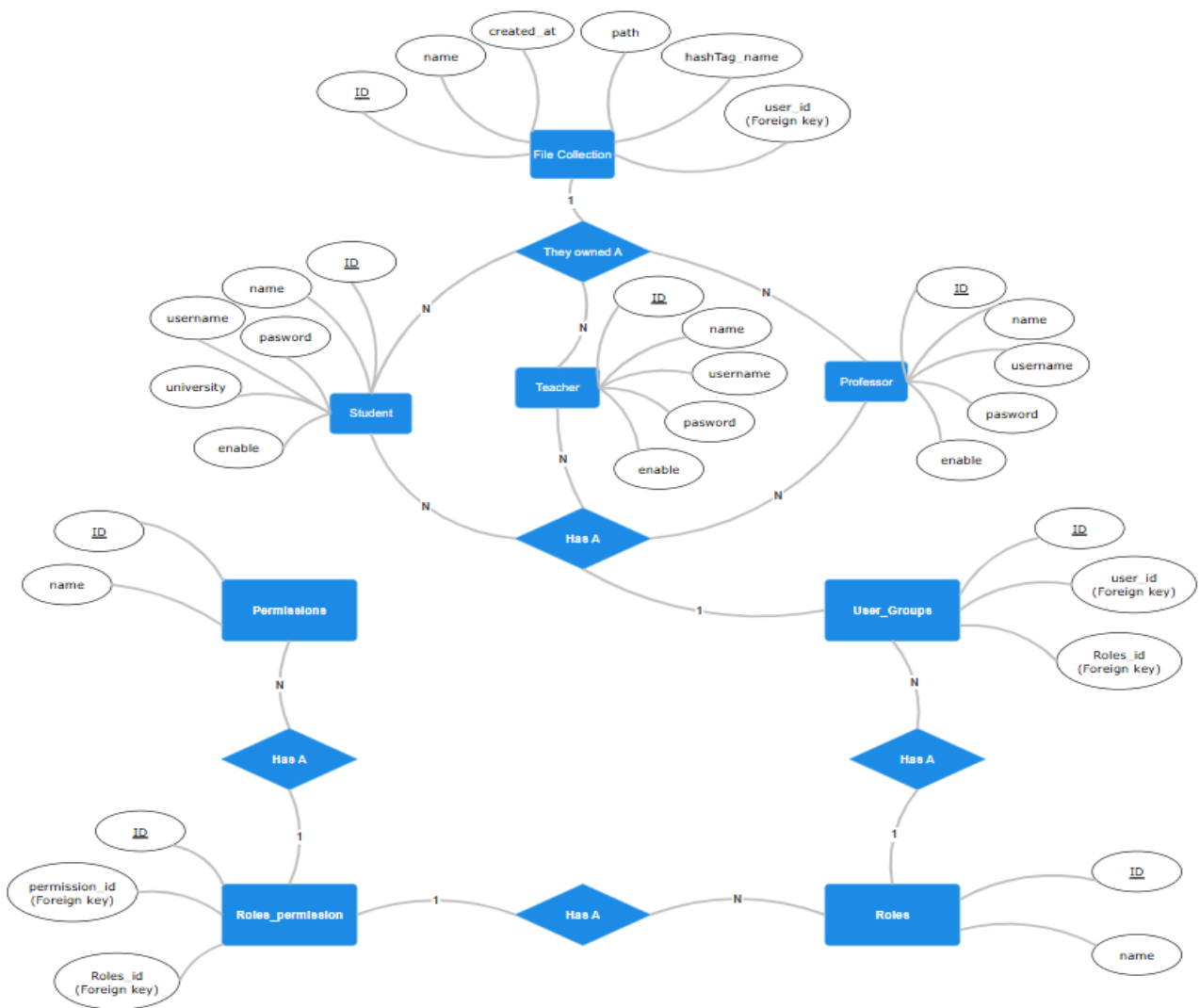


Figure 11 (4.3): ER Diagram

4.4 User interface design (prototype)

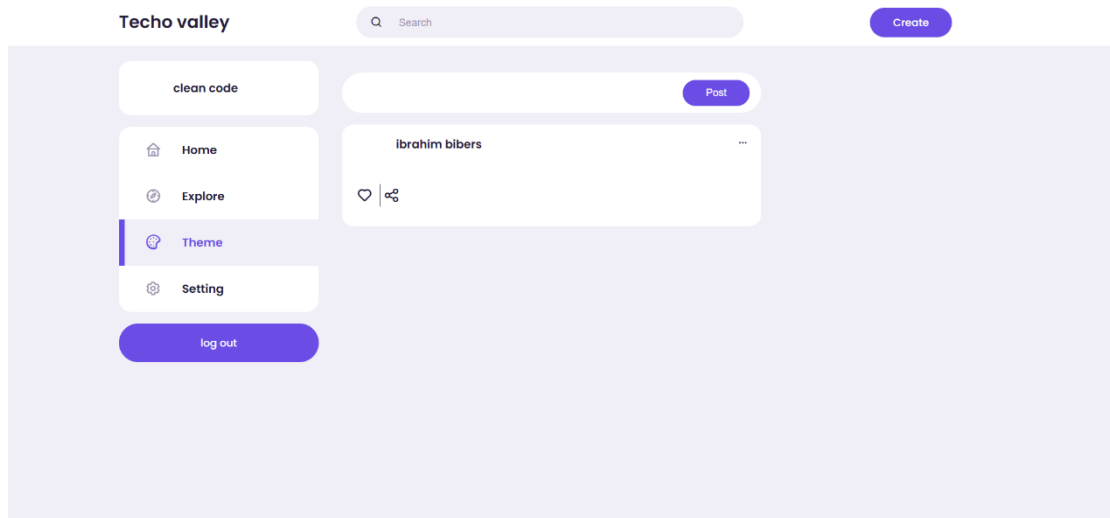


Figure 12 (4.4): User interface

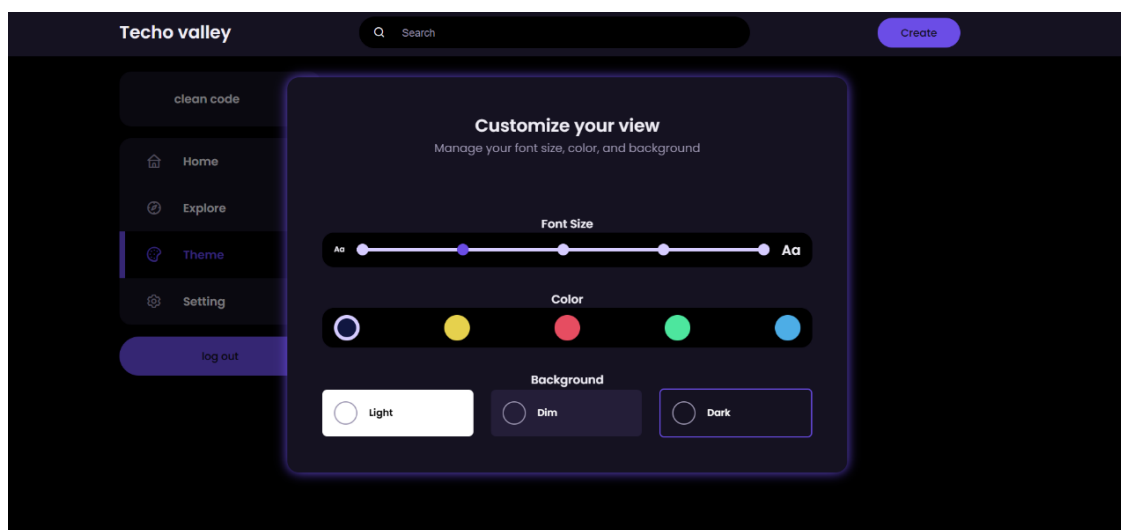


Figure 13 (4.5): User interface

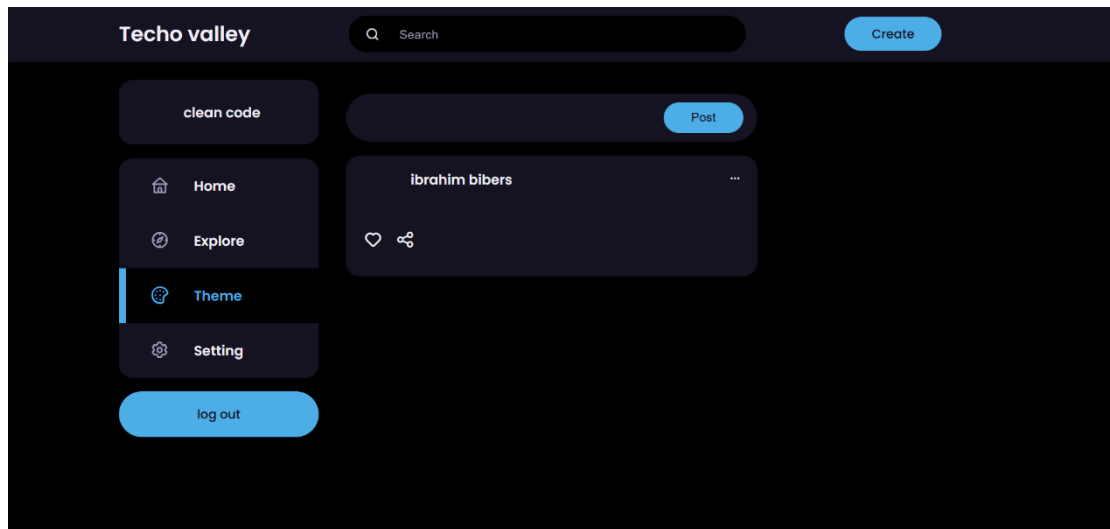


Figure 14 (4.6): User interface

CONCLUSION AND FUTURE DIRECTION

Conclusion:

Techno-Valley successfully addresses the need for a dedicated academic platform for IT faculties in Jordanian universities. The platform provides a structured environment for collaboration by enabling users to publish and access academic content with advanced search functionality. With role-based access control and a hashtag-based system, the platform ensures both content relevance and user security. The modular architecture and scalable design make it adaptable to future needs. Initial testing confirms the system's usability and performance, offering a reliable tool for students and academics.

Future Directions:

The interplay between social academic environments and technical projects is expected to evolve with advancements in technology and changes in educational paradigms. Future research should focus on designing inclusive and sustainable academic models that leverage global collaboration and innovation.

While the platform meets its current objectives, several enhancements are planned for future iterations:

1. **Integration of Video Content:** Support for video lectures and tutorials to enrich the learning experience.
2. **Advanced Analytics:** Implementing tools to track user engagement and popular topics.
3. **Mobile App Development:** Launching a dedicated mobile application for better accessibility.
4. **AI-Powered Search:** Enhancing the search functionality with AI to deliver more precise and context-aware results.
5. **Expanded User Roles:** Introducing additional roles, such as alumni and industry experts, to foster wider collaboration.

Sincere gratitude is extended to Dr. Khaled Awad for his invaluable support and guidance throughout the project. His insights and encouragement were instrumental in its success. His dedication and commitment to advancing academic collaboration in IT are deeply appreciated.

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